

Stoichiometric Calculations

1. How many grams of lithium must be reacted with oxygen to obtain 60 g of lithium oxide ?
2. How many grams of limestone (calcium carbonate) must be decomposed to obtain 448 L of carbon dioxide gas at STP?
3. How many liters of oxygen gas can be produced at STP from the decomposition of 2 moles of potassium chlorate?
4. How many grams of hydrochloric acid must be used to neutralize 10 g of sodium hydroxide?
5. How many grams of potassium must be dissolved in water to obtain 33,6 L of hydrogen gas at STP?
6. A 50 g sample of calcium carbonate (limestone) decomposes by heating.
 - a) How many molecules of calcium oxide can form?
 - b) How many liters of carbon dioxide gas can form at STP?
7. How many grams of aluminum must be reacted with sulphuric acid to obtain 11,2 L of hydrogen gas at STP?
8. How many grams of barium hydroxide must be used to neutralize 7,3 g of hydrochloric acid?
9. How many liters of hydrogen and oxygen gases can be produced from the electrolysis of 3,6 g of water?
10. How many liters of oxygen are needed to burn 32 g of methane (CH_4) completely?

- 11.** A 4,6 g sample of ethyl alcohol is burned.
- a)** How many water molecules can be formed?
 - b)** How many liters of carbon dioxide can be formed at STP?
- 12.** When a 20 g sample of calcium carbonate is reacted with excess hydrochloric acid, how many liters of carbon dioxide will form at STP?
- 13.** How many liters of nitrogen and hydrogen must be reacted with each other to obtain 20 L of ammonia?
- 14.** How many grams of sodium hydroxide are needed to neutralize 4,9 g of sulphuric acid?
- 15.** How many liters of hydrogen gas can evolve at STP from the reaction of 5,4 g of aluminum with excess hydrochloric acid solution?
- 16.** How many liters and grams of oxygen gas can be produced from the electrolysis of 9 g of water at STP?
- 17.** How many liters of oxygen are liberated by heating of 490 g of potassium chlorate at STP?
- 18.** Calculate the volume of air required to burn 4,6 of ethyl alcohol at STP.